Anup Raj Niroula

a.r.niroula@gmail.com | linkedin.com/in/anup-raj-niroula | github.com/ARNiroula | arniroula.github.io

Education

New York University — New York, USA
Master of Science in Computer Engineering — *September 2024 – May 2026*

Kathmandu University — Bagmati, Nepal
Bachelor of Engineering in Computer Engineering — *August 2017 – March 2022*

Experience

Graduate Teaching Assistant — January 2025 – Present
New York University — *New York, USA*

- Facilitated office hours for Machine Learning course to answer questions, explain concepts, and provide additional support to students
- Revised machine learning assignments by integrating real-world datasets and clearer instructions, leading to a 20% improvement in student scores and positive feedback

Artificial Intelligence Engineer — May 2025 – August 2025
MedLaunch Concepts — *Florida, USA*

- Built an event-driven Policy Analyzer using AWS S3, Bedrock, Lambda, and Step Functions to automate parallel analysis of hospital policies during accreditation updates
- Engineered a highly scalable, serverless ETL data pipeline for document conversion with AWS S3, Bedrock, and Lambda, consistently managing 1,000+ concurrent parses
- Integrated an AI-powered chatbot using Retrieval-Augmented Generation (RAG) and vector database, reducing manual query response time by 80%

Software Engineer — October 2021 – May 2023
Nimble Clinical Research — *New Jersey, USA*

- Led the R Programming Team and managed a team of 6 people
- Developed and maintained RESTful API communication between Django and R microservices
- Engineered Report Automation System, which generated reports of over 100,000 rows within a minute. Reduced report generation time by making it approximately 9x faster
- Devised an automated unit testing framework to run test suites during the CI stage, ensuring code quality and early bug detection
- Trained a supervised learning model for automated clinical data mapping, achieving 70%+ accuracy, improving efficiency in data standardization for clinical trials
- Optimized the codebase, reducing server startup time in the Kubernetes cluster by 5×

Projects

Spotify Buddies | *MLflow, FastAPI, PostgreSQL, Redis, Grafana, Airflow, Docker* — GitHub Link

- Trained and deployed a collaborative filtering recommendation model on Chameleon Cloud, generating up to 200,000 predictions per minute while maintaining high accuracy
- Redesigned the deployment pipeline, reducing model server launch time from 300s to 30s, and keeping the server image size under 2 GB
- Integrated real-time monitoring using Grafana and Prometheus to visualize model performance metrics and detect cold-start scenarios
- Automated retraining and model management using MLflow and Apache Airflow, improving system scalability and maintenance
- Developed multiple RESTful APIs using FastAPI and built an intuitive user interface to enable user interaction and feedback collection, with feedback data stored in PostgreSQL

News Classification (LoRA Fine-Tuning) | *PyTorch, Transformers, Tensorboard* — GitHub Link

- Fine-tuned LLMs using LoRA on news datasets with HuggingFace Transformers
- Applied domain adaptation techniques to enhance performance on target-specific news data
- Achieved test accuracy of more than 93%
- Secured 1st place in an internal Kaggle-style competition on LoRA-based fine-tuning organized by NYU

Technical Skills

Languages: Python, Go, R, JavaScript/TypeScript, SQL (Postgres, MySQL, SQLite), C/C++, Bash
Cloud Platforms: Amazon Web Services, Google Cloud Platform, Chameleon Cloud
Frameworks: FastAPI, Django, Flask, React, NextJS, Astro, Node.js
Libraries: Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, Tidyverse, data.table
Developer Tools: Git, Docker, Kubernetes, Argo CD, VS Code, Neovim, Vim